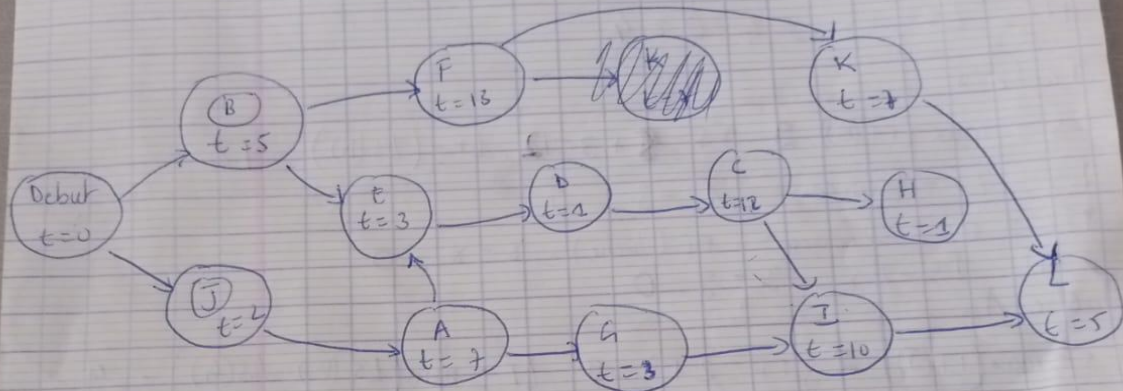


Exercice monôme TD : Diagramme de Gantt

Exercice

1- Complétons le diagramme de réseau



2- Calculons les marges totales et libres

ES : Date de début au plus tôt, EF : Date de fin au plus tôt

LS : Date de début au plus tard, LF : Date de fin au plus tard

MT : Marge totale ; ML : Marge libre

Task J: ES (min tous les successeurs)

$$ES = 0, EF = 0 + 2 = 2, LF = LS(A) = 2, LS = LF - \text{Durée} = 2 - 2 = 0$$

$$MT = LS - ES = LF - EF = 0 - 0 = 0 = 2 - 2 = 0$$

$$ML = ES(\text{successeur}) - EF(\text{tâche}) = 2 - 2 = 0$$

$$ES = (\max EF \text{ successeurs}) - \text{précédent}$$

Tache B

$$ES = 0, EF = 0 + 5 = 5, LF = \min(LS(F), LS(E)) = 9$$

$$LS = LF - \text{Durée} = 9 - 5 = 4$$

$$MT = LS - ES = 4 - 0 = 4$$

Tache E

$$ES = \max(EF(B)) = 5, EF = 5 + 3 = 8, LF = (LS(D)) = 12$$

$$LS = LF - \text{Durée} = 12 - 3 = 9$$

$$ML = ES(\text{successeur}) - EF(\text{tache}) = 8 - 8 = 0$$

$$MT = LS - ES = 9 - 9 = 0$$

Tache A:

$$ES = \max(EF(J)) = 2, EF = 2 + 7 = 9, LF = \min(LS(E), LS(G)) = 9$$

$$LS = LF - \text{Durée} = 9 - 7 = 2, MT = LS - ES = 2 - 2 = 0$$

Tache F

$$ES = \max(EF(B)) = 5, EF = 5 + 13 = 18, LF = \min(LS(K)) = 28$$

$$LS = LF - \text{Durée} = 28 - 13 = 15, MT = LS - ES = 15 - 5 = 10$$

Tache D

$$ES = \max(EF(E)) = 8, EF = 8 + 1 = 9, LF = \min(LS(L)) = 13$$

$$LS = LF - \text{Durée} = 13 - 1 = 12, MT = 12 - 9 = 3$$

Tache G

$$ES = \max(EF(A)) = 9, EF = 9 + 3 = 12, LF = \min(LS(I)) = 25$$

$$LS = LF - \text{Durée} = 25 - 3 = 22, MT = 22 - 9 = 13$$

Tâche K:

$$ES = \max(EF(F)) = 18, EF = 18 + 7 = 25, LF = \min(LS(L)) = 35$$

$$LS = 20, LS = LF - 7 = 35 - 7 = 28$$

$$MT = 28 - 18 = 10$$

Tâche L:

$$ES = \max(EF(K), EF(I)) = 35, EF = 35 + 5 = 40$$

$$LF = 40, LS = LF - 5 = 40 - 5 = 35, MT = LS - ES = 35 - 35 = 0$$

Tâche H:

$$ES = \max(EF(C)) = 25, EF = 25 + 1 = 26, LF = 40$$

$$LS = 40 - 1 = 39, MT = 39 - 25 = 14$$

Tâche C:

$$ES = \max(EF(D)) = 13, EF = 13 + 12 = 25, LF = \min(LS(H), LS(I)) = 25$$

$$LS = 25 - 12 = 13, MT = 13 - 13 = 0$$

Tâche I:

$$ES = \max(EF(G), EF(C)) = 25, EF = 25 + 10 = 35$$

$$LF = \min(LS(L)) = 35, LS = 35 - 10 = 25, MT = 25 - 25 = 0$$

3) Chemin critique

J → A → E → D → C → I → L

$$2 + 7 + 3 + 1 + 12 + 10 + 5 = 40 \text{ semaines}$$

4) Durée projet = Date de fin au plus tôt des tâches

terminal : $\begin{matrix} 26 & 40 \\ H & L \end{matrix} = 40$ semaines

On prend le plus grand.